

The price improvement optimization is:

$$\text{Max: } \text{Prob}(\text{Cost} \leq C^*) \quad (8.24a)$$

where C^* is the specified target price, cost, or principal bid that the investor is seeking to outperform. Mathematically, this optimization can also be written as:

$$\text{Max: } \frac{C^* - E[\text{Cost}]}{TR'} \quad (8.24b)$$

Example: An investor seeking to maximize the probability of achieving price improvement over a market impact cost of C^* (not including price appreciation cost) will determine the optimal trading rate via the following optimization:

$$\text{Max: } \frac{C^* - (b_1 \cdot I' \cdot \alpha + (1 - b_1) \cdot I')}{\sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot \frac{X}{ADV} \cdot \alpha^{-1} \cdot P_0}} \quad (8.25)$$

Solving for the optimal trading rate we get:

$$\alpha^* = \frac{C^* - K}{3 \cdot b_1 \cdot I} \quad (8.26)$$

Figure 8.2e illustrates the process used to determine the price improvement strategy for a cost of \$0.30/share. The top graph shows the efficient trading frontier with cost on the y-axis and timing risk on the x-axis. The bottom graph shows the probability that each of the strategies on the efficient trading frontier will incur a cost of less than \$0.30/share. The probability was determined assuming a normal distribution with mean equal to the expected cost (y-axis) and standard deviation equal to the timing risk (x-axis). The strategy that maximizes the likelihood that the cost will be less than \$0.30/share is found by drawing a line from the cost of \$0.30 on the y-axis tangent to the efficient trading frontier. This is denoted by strategy A5 and has expected cost of \$0.18/share and timing risk of \$0.40/share. The strategy has a 62.1% chance of outperforming \$0.30/share. Strategy A6 has an expected cost of \$0.30/share and timing risk of \$0.23/share and so a probability of 50% of outperforming \$0.30/share. Obviously, any strategy to the left of A6 will have an expected cost higher than \$0.30/share so the probability that the actual cost will be less than \$0.30/share will be less than 50%. Strategy A7 has an expected cost of \$0.10/share and timing risk of \$0.84/share. The corresponding probability that this strategy will have a cost less than \$0.30/share is 59.7%. Notice the shape of this probability curve. The probability of outperforming \$0.30/share increases fast until strategy A5 (the